Stat 560 project: presentation & report

Topic choice:

- research-adjacent
- topic of interest

(ask me!)

Presentation: during one lecture: $\frac{80\text{ min} - \text{breaks}}{5} \approx 15$ min

$= \underbrace{12}_{\text{talk}} + \underbrace{3}_{\text{q \& switch.}}$ min

Teach us something!

- lesson on topic. (do something!)
- investigation (numerical simulations, proof, data analysis [fit to Markov chain]) & report.

Report: ~8 pgs not including references.
(10 pgs if several figs.).

☆ Today: chapter 3, renewal processes

☆ Definition: let (t_n) be i.i.d. sequence, $t_n > 0$ a.s. $\forall n$.

Let $T_n = t_1 + \cdots + t_n$.

Let $N(t) = \max\{n : T_n \le t\}$.

$N(t)$ = # "points" in $[0, T]$ where "points" = $\{T_n\}_{n \ge 1}$.

if $t_n \overset{d}{=} \exp(\lambda)$, $N(t)$ is Poisson process w/ intensity λ.

$N(t)$: renewal process associated to waiting times (t_n).

<u>Note</u>: Poisson process has independent increments:

If $s_0 < s_1 \cdots < s_n$ are fixed,

$$\left(N(s_{i+1}) - N(s_i)\right)_{i=0}^{n-1} \quad \text{independent}$$

This property uses memoryless property of exponential.
Don't care when most recent point was when determining additional time until next point.

Not true for general renewal process!

Ex: return time sequence for discrete time Markov chain

fix $x \in S$, let $T_x^0 = 0$

$$T_x^k = \inf\{n > T_x^{k-1} : X_n = x\}$$

$$N(n) := \max\{k : T_x^k \le n\}$$

visits to x in $[0,n]$
(minus initial visit if $X_0 = x$).

Renewal theorem:

if $E t_i < \infty$ then

$$\lim_{t \to \infty} N(t)/t = 1/E[t_i] \quad \text{a.s.}$$

events in $[0,t] = t \cdot$ frequency

$= t/\text{waiting time}$

Can be proved from strong law of large #'s, that says

$$T_n/n \xrightarrow{n \to \infty} E[t_i] \quad \text{a.s.} \qquad (*)$$

Idea:

$$N(t) = n \quad \forall \quad T_n \le t < T_{n+1}$$

for such t

$$\frac{T_n}{n} = \frac{T_n}{N(t)} \le \frac{t}{N(t)} \le \frac{T_{n+1}}{N(t)} = \frac{T_{n+1}}{n+1} \cdot \frac{n+1}{n}$$

as $t \to \infty$, $n \to \infty$. Then use $(*)$.

Mild extension: renewal-reward process

Suppose each time T_i
we get a "reward" r_i
drawn from i.i.d. sequence.

Let $$R(t) = \sum_{i=1}^{N(t)} r_i$$ reward to time t.

Then $$R(t)/t \xrightarrow{t \to \infty} E[r_i]/E[t_i]$$

why?

$$\frac{1}{E[t_i]} = \lim_{t \to \infty} N(t)/t$$

$$E[r_i] = \lim_{n \to \infty} 1/n \sum_{i=1}^{n} r_i .$$

example: alternating renewal process:

pair of sequences r_i, s_i

$$t_i = r_i + s_i$$

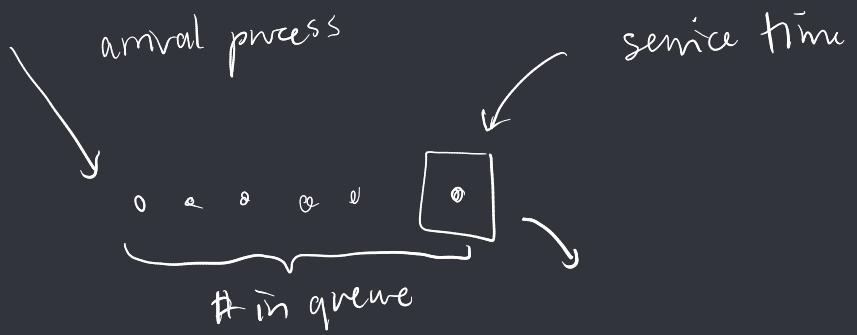
r_i = time machine working

s_i = service/repair time

fraction of time working long-term

$$= \frac{E[r_i]}{E[r_i] + E[s_i]}$$

<u>Application</u>: queuing theory



assume arrivals occur according to renewal process (i.i.d. times between arrivals)

& service times are i.i.d.

Interested in long-term behavior & averages.

Waiting times (t_i) $\quad E[t_i] = 1/\lambda$

service times (s_i) $\quad E[s_i] = 1/\mu$

λ : arrival rate

μ : service rate

Under what conditions does queue remain finite?

If $\lambda < \mu$ expect queue to be empty infinitely often.

if $\lambda > \mu$ expect queue size $\nearrow \infty$.

if $\lambda = \mu$ ... ?

Fraction of time queue is busy $= \lambda/\mu$

$(\to 1$ as $\lambda \nearrow \mu^-)$

(to see queue becomes empty inf-often
compare S_n/n to T_n/n

where $S_n = s_1 + \cdots + s_n$
$T_n = t_1 + \cdots + t_n$))

Formulas connecting average quantities in queue.
Let

$$L = \lim_{t \to \infty} \frac{1}{t} \int_0^t X_s \, ds$$

avg. # in system over time

X_s = # individuals in system @ time s.

$$W = \lim_{n \to \infty} \frac{1}{n} \sum_{m=1}^{n} W_m$$

W_m = time spent by m^{th} individual in system.

Then $$L = \lambda W.$$

avg. size = (arrival rate) · (avg time spent).

<u>Proof</u>: imagine 2 ways to count:

- each individual gives you \$1 / minute spent in queue.

- each individual (m^{th} individual) gives you \$$W_m$ when they arrive.

In both, in long run the rate of profit is same.

Connecting queue size to service time gives proportion of time queue is empty:

let $W_Q :=$ E[time] spent in queue (but counting service time).

then $W_Q = W - E[s_i]$.

Similar reasoning shows $L_Q = \lambda W_Q$

$L_Q :=$ average # in queue (not counting served)

and if $\pi(0) :=$ fraction of time: no one in system.

then $L_Q = L - 1 + \pi(0)$

in queue = # in system − 1

except when
no one is there.

using formulas,

$$\lambda W_Q = \lambda W - 1 + \pi(0)$$

$$1 + \lambda \underbrace{(W_Q - W)}_{= -E[s_i]} = \pi(0)$$

$$\therefore \pi(0) = 1 - \lambda E[s_i]$$

$$= 1 - \lambda/\mu.$$

Age & residual life

wait for a large fixed time t.

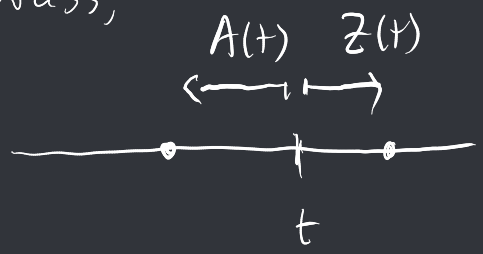
examine time since most recent event (1)

& time until next event. (2)

(1) $A(t) := t - T_{N(t)}$ age

(2) $Z(t) = T_{N(t)+1} - t$ residual life.

for Poisson process,



for any t,

$A(t)$ & $Z(t)$

are indep. $\exp(\lambda)$.

for general renewal process, marginal dist$^{\underline{bn}}$. of $A(t)$, $Z(t)$ approaches (as $t \to \infty$) size-biased dist$^{\underline{bn}}$. of waiting time.

size-biasing: if $X \in \mathbb{N}$ w/ $P(X=k) = p_k$

& $E[X] =: m < \infty$ then

size-biased dist'bn $\tilde{p}$

has

$$\tilde{P}(X=k) = k\, p_k / m.$$

more generally if $X \geq 0$, size-biased $\tilde{p}$ has

$$\tilde{P}(X \in A) = \int_{x \in A} x \, dP(x) \Big/ E[X].$$

why size-biased? (for $A(t)$, $Z(t)$)

? If "large fixed t" gives same distbtn.
as "$t \overset{d}{=} \text{unif}[0, T]$"

then for large T, probab. to choose an interval is proportional to its length.

why ? would these distbtns. be the same?
under what conditions?

Analogous to frequency of visits vs. convergence of transition probabilities

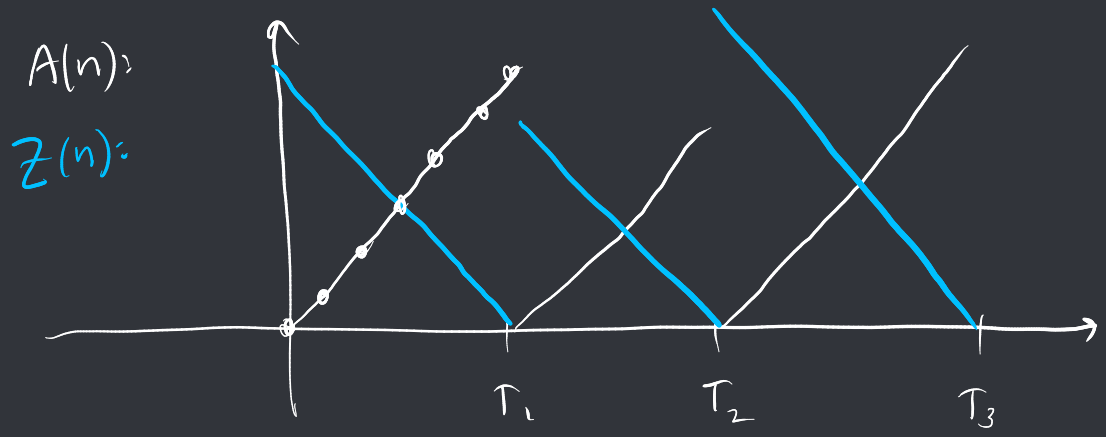
$$\lim_{n\to\infty} \frac{1}{n} \#\left(\text{visits to } x \text{ in } [0,n]\right)$$

$$p^n(x,y) \to \pi(y)$$
$$n \to \infty$$

Discrete time analysis

suppose $T_n \in \mathbb{N}$ $\forall n$.

Then



$(A(n))_{n\in\mathbb{N}}$ increases by 1 @ each step, resets at each time T_n.

$(Z(n))_{n\in\mathbb{N}}$ jumps by $t_{n+1}-1$ at time T_n+1, decreases to 0.

We'll study $Z(n)$. It is a Markov chain with

$$p(0,j) = P(t_i = j+1)$$

$$p(j,j-1) = 1, \quad j > 0$$

what is stat. dist.? Look at excursions from 0.

$$\pi(j) = \frac{E_0[\#\text{visits to } j \text{ before back to } 0]^{(*)}}{\underbrace{E_0[\text{return time to } 0]}_{= E[t_i]}}$$

$$(*) = P(t_i \geq j+1)$$

1 visit if jump to $k \geq j$

0 visits else.

$$\pi(j) = P(t_i > j) \Big/ E[t_i]$$

for dist$^{\underline{bn}}$ of $Z(n)$ to converge as $n \to \infty$ enough if

- $E[t_i] < \infty$
- process irreducible (true if $P(t_i > j) > 0 \quad \forall j$),
- process aperiodic (true if $\gcd(\{j : p(0,j-1) > 0\}) = 1$)

how does dist$^{\underline{bn}}$ compare to original?

$$E[Z(n)] \text{ for large } n \approx \sum_j j\,\pi(j)$$

$$= \sum_j j \frac{P(t_i > j)}{E[t_i]}$$

$$\left(\vdots\right)$$